

– SAFETY DATA SHEET

NN-RockGrout Resin Capsule

Resin Capsule – Two-Component Polyester Grouting System

Prepared in the format of GHS / the Chemicals (MSIHC) Rules

Document No.	Revision	Issue Date	Product Code
NNPL/SDS/PROD-04	1.0	29 June 2026	RC-04

01 IDENTIFICATION

Field	Detail
Product name	NN-RockGrout Resin Capsule
Product type	Two-component polyester resin grouting capsule
Recommended use	Anchorage / grouting medium for roof & rock bolting (underground)
Supplier	NilayNarayan Polychem LLP, Plot D-167 (P), Kandra Ind. Area, Govindpur, Jharkhand 826010
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02 HAZARD IDENTIFICATION

The capsule contains two reactive components. Hazards relate mainly to the styrene-bearing resin and the organic-peroxide catalyst; the cured grout is an inert solid.

- **Flammable** – Resin component (styrene): flammable liquid and vapour.
- **Oxidiser** – Catalyst (organic peroxide): heating may cause fire; strong oxidiser.
- **Irritant** – Causes skin and serious eye irritation; may cause skin sensitisation.
- **Vapour** – Vapour may cause respiratory irritation and drowsiness or dizziness.
- **Health** – Suspected of causing harm on prolonged styrene exposure; handle with care.

Precautionary summary: keep away from heat, sparks and open flame. Avoid breathing vapours. Wear gloves and eye protection. Keep resin and catalyst separated until use.

03 COMPOSITION / INFORMATION ON INGREDIENTS

Ingredient	CAS No.	Remark
Unsaturated polyester resin	—	Resin matrix base
Styrene (reactive diluent)	100-42-5	Flammable; irritant; reprotoxic susp.
Mineral fillers (carbonate / silicate)	—	Inert filler
Organic peroxide (benzoyl peroxide)	94-36-0	Catalyst; oxidiser; irritant

04 FIRST-AID MEASURES

- **Inhalation** – Move to fresh air; keep at rest. If breathing is difficult, get medical attention.
- **Skin** – Remove contaminated clothing; wash skin with soap and water. If irritation or sensitisation persists, seek medical advice.
- **Eyes** – Rinse cautiously with water for several minutes; remove contact lenses if easy. Seek medical attention.

- **Ingestion** — Do not induce vomiting; rinse mouth; obtain medical attention.

05 FIRE-FIGHTING MEASURES

- **Suitable media** — Dry chemical powder, CO₂, alcohol-resistant foam or water spray.
- **Hazards** — Styrene vapour is flammable; peroxide decomposition accelerates combustion and may cause violent burning.
- **Protection** — Self-contained breathing apparatus and full protective gear for fire-fighters.

06 ACCIDENTAL RELEASE MEASURES

- **Safety** — Eliminate ignition sources; ventilate the area; avoid contact and vapour inhalation.
- **Containment** — Do not allow resin and catalyst to combine uncontrolled — exothermic reaction.
- **Clean-up** — Absorb with inert material; collect into suitable containers; dispose per local regulations.

07 HANDLING & STORAGE

- **Handling** — Handle capsules gently to avoid rupture; wear gloves and eye protection.
- **Separation** — Keep resin and catalyst components separated; protect from heat, sparks and direct sunlight.
- **Storage** — Store cool, dry and shaded, away from oxidisers, acids and accelerators; protect from extreme weather.
- **Stock** — Rotate stock; observe shelf life and re-test gel/set time of the oldest batch monthly.

08 EXPOSURE CONTROLS / PERSONAL PROTECTION

Control	Recommendation
Styrene — occupational exposure	Observe applicable national OEL (verify locally)
Respiratory protection	Organic-vapour respirator where ventilation is inadequate
Hand protection	Nitrile / chemical-resistant gloves
Eye protection	Safety goggles / face shield
Engineering controls	Adequate underground ventilation; local exhaust where practicable

09 PHYSICAL & CHEMICAL PROPERTIES

Property	Value
Appearance	Putty-like mastic in flexible film cartridge
Odour	Characteristic (styrene)
Physical state	Semi-solid paste (two components)
Max reaction (cure) temperature	≤ 80 °C
Flammability	Resin component flammable; cured resin very low flammability
Solubility in water	Insoluble (cures to inert solid)

10 STABILITY & REACTIVITY

- **Stability** — Stable under recommended storage; components react when mixed (designed exotherm).
- **Avoid** — Heat, ignition sources and premature mixing of resin and catalyst.
- **Incompatibles** — Oxidisers, acids, bases, accelerators (cobalt salts, amines) and reducing agents.
- **Decomposition** — CO, CO₂ and irritant aromatic vapours on combustion.

11 TOXICOLOGICAL INFORMATION

- **Styrene** — Irritant to skin, eyes and respiratory tract; CNS depressant at high vapour levels; suspected reproductive toxicant on prolonged exposure.
- **Benzoyl peroxide** — Irritant and skin sensitiser; oxidiser.
- **Cured resin** — Inert solid, low hazard under normal handling.

12 ECOLOGICAL INFORMATION

Uncured components may be harmful to aquatic life; styrene is volatile and biodegradable. Prevent release of uncured material to soil, drains and watercourses. Cured resin is inert.

13 DISPOSAL CONSIDERATIONS

Allow waste resin and catalyst to fully cure where safe, then dispose of the cured solid as inert waste per local regulation. Uncured waste and contaminated packaging are handled as hazardous waste through authorised handlers.

14 TRANSPORT INFORMATION

Classify per applicable transport rules for the components (organic-peroxide catalyst and flammable styrene-bearing resin). Consult current dangerous-goods regulations for UN number, packing group and labelling appropriate to the packaged form.

15 REGULATORY INFORMATION

Product manufactured and supplied to meet DGMS Circular (Approval) No. 04 of 2013 standards for resin capsules used as roof-bolting grouting medium in belowground coal mines. Handle in accordance with applicable chemical-safety and mines regulations.

16 OTHER INFORMATION

This SDS is provided for safe-handling guidance. Exposure limits and transport classifications must be verified against the regulations in force at the place of use. Issued by the NilayNarayan Polychem LLP NABL-accredited laboratory.

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